

5(a)(i)	light intensity has maximum value at 0° , 180° , 360° and zero intensity at 90° , 270°	M1
	'sinusoidally-shaped' curve	A1
5(a)(ii)	$4.2 = 7.6 \cos^2 \theta$	C1
	$\theta = 42^\circ$	A1
5(b)	wave passes (through) an aperture or wave passes (by / through / around) an edge	B1
	wave spreads (into geometrical shadow)	B1
5(c)(i)	$n\lambda = d \sin \theta$	C1
	$d = (3 \times 4.3 \times 10^{-7}) / \sin 68^\circ$ $= 1.4 \times 10^{-6} \text{ m}$	A1
5(c)(ii)	$1.4 \times 10^{-6} \times \sin 68^\circ = 2 \times \lambda$ or $3 \times 4.3 \times 10^{-7} = 2 \times \lambda$	C1
	$\lambda = 6.5 \times 10^{-7} \text{ m}$	A1